

Agentic Lakehouse Growth Engine

Intelligent Marketing Agent

Powered by watsonx.data Lakehouse · MCP Tool Bridge

The Problem

Why marketing teams are stuck



Data locked in silos

Customer data lives deep inside watsonx.data lakehouse tables — inaccessible without SQL expertise or BI tooling.



Generic campaigns

Marketers send the same email to all customers. No segmentation, no personalization, no relevance — low conversion.



Manual, slow pipelines

Pulling purchase history, profiling customers, writing emails — done manually. Hours of work per campaign cycle.



No real-time intelligence

Insights are weeks old by the time they reach the marketing team. Customer behavior changes faster than reports do.

Agentic Lakehouse Growth Engine

One AI agent. Real data. Personalized action.

Agentic Lakehouse connects your marketing team directly to live customer data in watsonx.data — and instantly generates personalized email campaigns without any SQL, BI tools, or manual effort.

1

Ask in plain English

No SQL. No data team needed.
Just type your question.



2

Agent queries live data

Pulls customers, transactions,
and product catalog in real-time.



3

Get campaign-ready output

Personalized email, discount code,
and product picks — instantly.

How It Works

The agent automatically detects query intent and routes to the right pipeline

MODE A

Direct Data Query

Fast · Single MCP call · No analysis

"list, show, describe"

Display raw data from any table

"count, how many"

Aggregate queries on the fly

"get, display"

Retrieve and format records

VS

MODE B

Campaign Generator

3-phase pipeline · Data + Analysis + Email

"recommend, suggest"

Triggers full 3-phase pipeline

"email, campaign, send"

Generates personalized outreach

"inactive, target, discount"

Customer segmentation + offer

Mode B Deep Dive

The 3-phase pipeline that turns a name into a campaign

Phase 01

Data Collection

- Get available Presto engine
- Look up customer by name or ID
- Fetch full customer profile
- Retrieve last 10 purchases
- Pull top-rated products ($\geq 4\star$)



Phase 02

Analysis

- Classify: Active vs Inactive
- Calculate lifetime value
- Identify preferred categories
- Select 3-4 matched products
- Calculate personalised discount



Phase 03

Email Generation

- Personalised subject line
- References past purchases
- Lists curated products
- Embeds discount code
- Includes call-to-action

Data Foundation

Three lakehouse tables powering every recommendation

watsonx.data Lakehouse · iceberg_data.mynta_sales_data · Presto Engine · Apache Iceberg Format

Customer Profiles

Key fields

customer_id (PK)

Name, age, Gender

city, state

Email

Profile enrichment, name-to-ID lookup, location context

Purchase History

Key fields

customer_id (FK)

product_id (FK)

date, original_price

discount

Activity scoring, category preferences, LTV calculation

Product Catalog

Key fields

product_id (PK)

product_name, category

brand_name, color, size

ratings (0–5)

Top-rated product selection matched to customer taste

Business Impact

< minute

Full campaign generated

vs hours manually

100%

Data-grounded output

no hallucinated records

2 modes

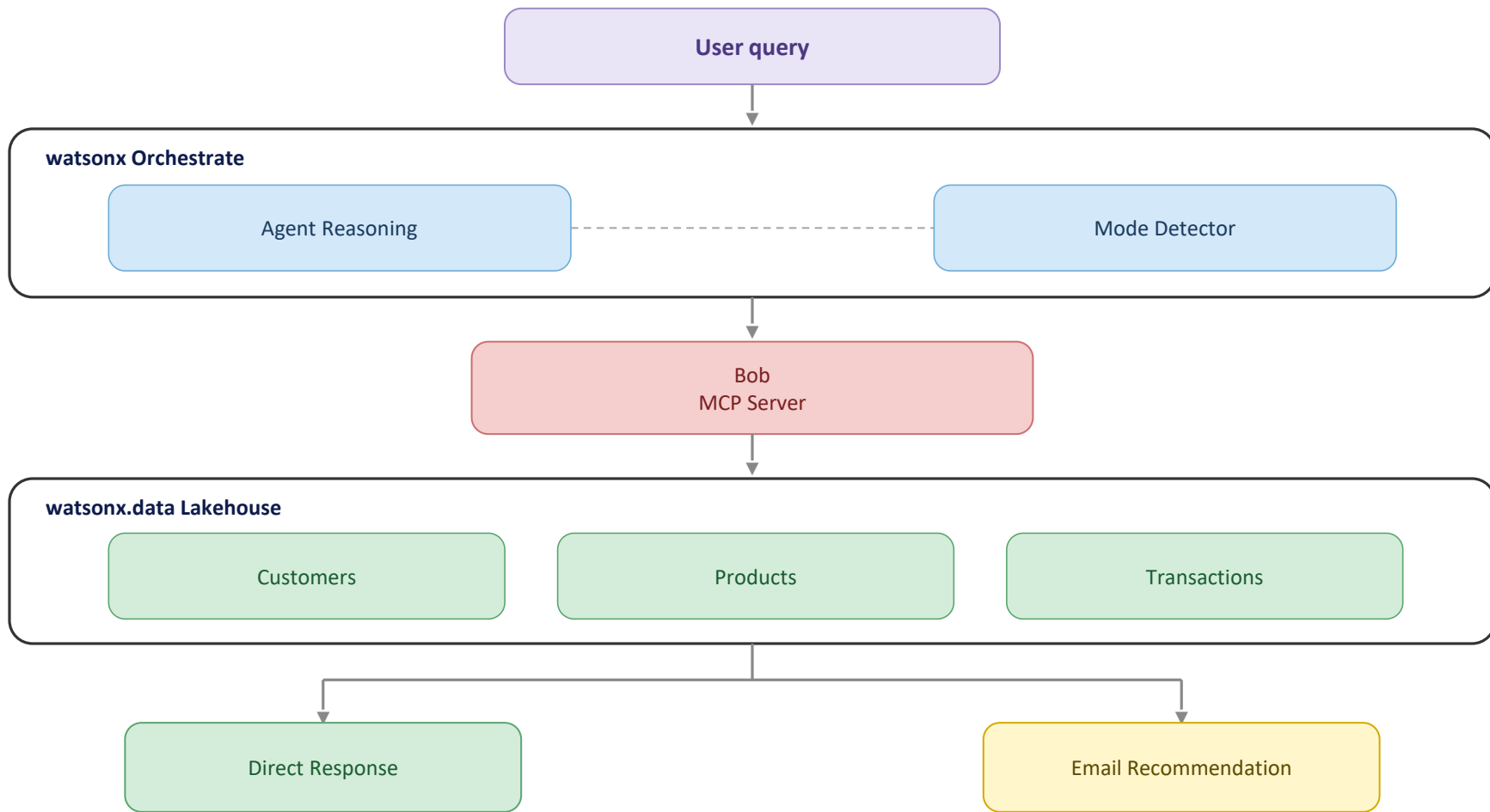
One agent, all use cases

queries + campaigns

tables

Live lakehouse joins

real-time intelligence



Lakehouse Growth Engine

Ask once. Get everything.

From a natural language prompt to a fully personalised email campaign — powered by live lakehouse data, IBM Granite LLM, and MCP tool orchestration.

watsonx.data
Apache Iceberg

watsonx Orchestrate

MCP Protocol